Assignment 8: Basic SQL Queries and Aggregate Functions

ProjectRapido – Bike & Cab Booking System
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1. Objective

Retrieve and summarise Rapido data using WHERE, ORDER BY, GROUP BY, HAVING and the aggregate functions COUNT, SUM, AVG, MIN and MAX. Each query is followed by the actual result set and a short explanation.

2. Filtering with WHERE

List only completed trips so operations can audit finished journeys.

```
SELECT ride_id, user_id, captain_id, pickup_location, drop_location, fare, status FROM rides
WHERE status = 'Completed';
```

ride_id	user_id	captain_id	pickup_location	drop_location	fare	status
1	1	1	Koramangala	HSR Layout	85	Completed
2	2	2	Indiranagar	MG Road	120	Completed
3	3	4	Whitefield	Koramangala	340	Completed
4	1	3	HSR Layout	MG Road	95	Completed
5	4	5	MG Road	Indiranagar	150	Completed

ride_id	user_id	captain_id	pickup_location	drop_location	fare	status
6	5	6	Whitefield	HSR Layout	280	Completed
10	8	5	Indiranagar	Whitefield	310	Completed
11	3	6	HSR Layout	Whitefield	250	Completed
12	4	1	Koramangala	MG Road	100	Completed
13	5	3	MG Road	Koramangala	80	Completed
14	6	2	Indiranagar	HSR Layout	75	Completed

Explanation: WHERE keeps 11 of 15 rides. Requested, ongoing and cancelled bookings are excluded.

3. Sorting with ORDER BY

Show all rides from highest fare to lowest (useful for long-distance / cab demand).

```
SELECT ride_id, pickup_location, drop_location, fare, status FROM rides ORDER BY fare DESC;
```

ride_id	pickup_location	drop_location	fare	status
8	MG Road	Whitefield	360	Requested
3	Whitefield	Koramangala	340	Completed
15	Whitefield	MG Road	330	Ongoing
10	Indiranagar	Whitefield	310	Completed

ride_id	pickup_location	drop_location	fare	status
6	Whitefield	HSR Layout	280	Completed
11	HSR Layout	Whitefield	250	Completed
5	MG Road	Indiranagar	150	Completed
2	Indiranagar	MG Road	120	Completed
12	Koramangala	MG Road	100	Completed
4	HSR Layout	MG Road	95	Completed
9	HSR Layout	Koramangala	90	Cancelled
1	Koramangala	HSR Layout	85	Completed
13	MG Road	Koramangala	80	Completed
14	Indiranagar	HSR Layout	75	Completed
7	Koramangala	Indiranagar	70	Ongoing

Explanation: The costliest booking is ride 8 (₹360, MG Road → Whitefield). The cheapest is ride 7 (₹70). ORDER BY does not hide rows; it only sorts them.

4. Grouping with GROUP BY

Count rides and total fare by life-cycle status.

```
SELECT status, COUNT(*) AS ride_count, ROUND(SUM(fare),2) AS total_fare FROM rides GROUP BY status;
```

status	ride_count	total_fare
Cancelled	1	90
Completed	11	1885
Ongoing	2	400
Requested	1	360

Explanation: Most business is completed (11 rides, ₹1885). GROUP BY collapses many ride rows into one row per status. COUNT(*) and SUM(fare) are applied inside each group.

5. Filtering groups with HAVING

Find captains whose completed-trip revenue exceeds ₹200.

```
SELECT c.captain_id, c.full_name, COUNT(r.ride_id) AS completed_rides, ROUND(SUM(r.fare),2) AS revenue FROM captains c JOIN rides r ON c.captain_id = r.captain_id WHERE r.status = 'Completed' GROUP BY c.captain_id, c.full_name HAVING SUM(r.fare) > 200 ORDER BY revenue DESC;
```

captain_id	full_name	completed_rides	revenue
6	Manoj Gowda	2	530
5	Lakshmi Bai	2	460
4	Deepak Joshi	1	340

Explanation: WHERE filters rows before grouping (completed only). HAVING filters after aggregation. Ravi Kumar's completed revenue is ₹185 (85+100), so he is correctly excluded even though he has trips.

6. Aggregate functions

6.1 COUNT, SUM, AVG, MIN, MAX on rides

```
SELECT COUNT(*) AS total_rides, SUM(CASE WHEN status = 'Completed' THEN 1 ELSE 0 END) AS completed_rides, ROUND(SUM(fare),2) AS sum_fare, ROUND(AVG(fare),2) AS avg_fare, MIN(fare) AS min_fare, MAX(fare) AS max_fare FROM rides;
```

total_rides	completed_rides	sum_fare	avg_fare	min_fare	max_fare
15	11	2735	182.33	70	360

Explanation: The platform has 15 bookings. Average advertised fare is ₹182.33, ranging from ₹70 to ₹360. COUNT(*) includes cancelled rides; the CASE expression counts completions separately.

6.2 AVG, MIN, MAX on captain ratings

```
SELECT ROUND(AVG(rating),2) AS avg_captain_rating, MIN(rating) AS min_rating, MAX(rating) AS max_rating FROM captains;
```

avg_captain_rating	min_rating	max_rating
4.52	4.2	4.8

Explanation: Fleet quality is high and tight: 4.20 to 4.80, mean 4.52.

6.3 COUNT and SUM on successful UPI payments

```
SELECT COUNT(*) AS successful_upi_payments, ROUND(SUM(amount),2) AS upi_amount FROM payments WHERE payment_method = 'UPI' AND payment_status = 'Success';
```

successful_upi_payments	upi_amount
5	680

Explanation: Five successful UPI collections total ₹680. The failed UPI row (ride 13) is excluded by the WHERE clause, which is the correct cash-position view.